

Ratios and Equivalent Ratio Tables

Grade 6–7 Mathematics

Writing ratios from models, part-to-part and part-to-whole forms, equivalent ratio tables, and the value of a ratio

Directions. Each problem gives you a model — a tile strip, a tape diagram, or a ratio table. Connect the model to notation: a ratio of a to b may be written a **to** b , $a : b$, or $\frac{a}{b}$. Show the model work you used, then write the answer on the line.

1. Tile strip. The strip below models one handful of tiles. Each R is a red tile and each B is a blue tile.

R	R	R	R	B	B	B	B	B	B
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Write the ratio of **red tiles to blue tiles** in all three forms, then give it in simplest form as a fraction.

Answer: _____

2. Same tile strip. Now compare a part with the whole handful. Write the ratio of **red tiles to all the tiles** as a fraction in simplest form, and say in words what that fraction tells you about the handful.

Answer: _____

3. Ratio table. A pancake recipe mixes flour and water in the ratio 3 to 2 (3 cups of flour for every 2 cups of water).

Water (cups)	2	4	6	8
Flour (cups)	3			

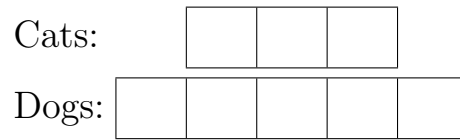
Complete the table. How many cups of flour go with 8 cups of water?

Answer: _____

4. Value of a ratio. A smoothie stand uses 12 strawberries for every 3 smoothies. Write the ratio of strawberries to smoothies, then find its **value**: the number of strawberries per single smoothie.

Answer: _____

5. Tape diagram. At a shelter the ratio of cats to dogs is 3 to 5. The tape diagram shows one box for each part, and every box stands for the same number of animals.



Each box stands for 4 animals. How many animals are at the shelter in all?

Answer: _____

6. Ratio table. Pencils are sold at a steady price: 5 pencils cost 2 dollars.

Pencils	5	10	20
Cost (dollars)	2		

Complete the table. What does a bundle of 20 pencils cost, in dollars?

Answer: _____

7. Part to whole. In one class, the ratio of students who walk to school to students who ride to school is $2 : 7$, and every student does exactly one of the two. What **fraction of the whole class** walks? Explain how the 7 in the ratio can turn into a 9 in your answer.

Answer: _____

8. Comparing by value. Two drink mixes are made with scoops of powder and cups of water.

Mix A: 4 scoops in 10 cups. Mix B: 6 scoops in 14 cups.

Write each mix as a ratio in fraction form, find the value of each ratio, then write $<$ or $>$ between them in this order: value of Mix A _____ value of Mix B.

Answer: _____

9. Ratio table, harder step. A lemonade recipe uses 9 lemons for every 6 cups of lemonade.

Cups of lemonade	6	2	14
Lemons	9		

Fill in the middle column first — it is the easy step — and use it to find how many lemons are needed for 14 cups.

Answer: _____

10. Synthesis. A mural paint is mixed from blue and yellow in the ratio 3 to 2. Nia needs 45 liters of mixed paint in total.

Draw a tape diagram with one box per part, work out what one box is worth, then find how many liters of **blue** paint Nia needs. State the ratio of blue paint to total paint as a fraction as part of your reasoning.

Answer: _____ L